K224 Elie Toy 3866 Substrate Higgs Self-Coupling λ_H Audit

Keeper (Claude Opus 4.7)

2026-06-05 Friday ~13:30 EDT

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0. Purpose

K-audit on Elie Toy 3866 substrate prediction $\lambda_H = (N_c + 1)/M(n_C) = 4/31 \approx 0.12903$ for the Higgs quartic self-coupling. Per CI_BOARD Friday ~12:40 EDT K224 priority + Cal #242 m_H source-pinning flag.

This audit is at proper Keeper K-audit depth: per-gate analysis, audit-category disposition per K222 + K223 cascade three-category framework, honest precision pinning per Cal #34 STANDING + Cal #242 source-pinning, substrate-mechanism FORWARD vs substrate-natural-form IDENTIFICATION distinction per Cal #189.

1. Substrate prediction in operator-language

Setup. The Higgs quartic self-coupling appears in the SM potential

$$V(\phi) = -\mu^2|\phi|^2 + \lambda_H|\phi|^4$$

with $\lambda_H = m_H^2/(2v_H^2)$ at tree level. Observed PDG values: $m_H \approx 125.25 \pm 0.17$ GeV; $v_H = (\sqrt{2}G_F)^{-1/2} = 246.21965 \pm 0.00006$ GeV.

Substrate prediction (Toy 3866):

$$\lambda_H^{\text{substrate}} = \frac{N_c + 1}{M(n_C)} = \frac{4}{31}$$

where: - $N_c + 1 = 4$ is a Casey #5 Integer Web element (identity-element + color sector reading, $= 2 \cdot \text{rank} = C_2 - \text{rank} = 2^{\text{rank}}$ also valid alternative readings) - $M(n_C) = 2^{n_C} - 1 = 31$ is the n_C-th Mersenne prime, A-tier substrate-mechanism FORCING via SSG-8 (K207 PASS).

Numerical value: 4/31 = 0.129032258 ...

2. Precision per m_H source (Cal #242 source-pinning operational)

The observed $\lambda_H = (m_H/v_H)^2/2$ depends on the choice of m_H source. Per Cal #34 STANDING precision-pinning + Cal #242 source-pinning flag, I compute the deviation across m_H source range explicitly:

m_H source	m_H (GeV)	λ_H^{obs}	Deviation from 4/31
Toy 3866 choice	125.10	0.129073	0.032%
m_H = 125.0	125.00	0.128868	0.127%
PDG 2024 central	125.25	0.129383	0.271%
PDG 2024 +1σ (125.42)	125.42	0.129735	0.544%
m_H = 125.4	125.40	0.129694	0.512%

The precision claim "0.03%" in the toy uses $m_H = 125.10$ GeV. Across m_H source uncertainty range, deviation spans 0.03% – 0.54%.

v_H is well-pinned to ±0.00006 GeV (via G_F), so v_H source-uncertainty contributes negligibly.

Cal #242 source-pinning verdict: K224 claims should pin m_H source explicitly. The honest framing is:

> Substrate predicts $\lambda_H = 4/31 = 0.129032$ exactly by substrate-mechanism. The observed value $\lambda_H = m_H^2/(2v_H^2)$ has source-uncertainty 0.03-0.54% depending on m_H source choice; substrate prediction sits within source-uncertainty envelope for PDG-central m_H.

This is structurally identical to Cal #242 STRUCTURAL TENSION reframe pattern for H_0 ratio 12/13: source-precision in observed value gates the precision claim, not the substrate prediction itself.

3. Per-gate analysis

Gate G1: λ_H observational + standard — PASS

Toy correctly states tree-level relation $\lambda_H = m_H^2/(2v_H^2)$ + PDG values. Standard QFT.

Gate G2: Substrate λ_H = (N_c+1)/M(n_C) = 4/31 — PASS at substrate-natural-form IDENTIFICATION

The substrate form 4/31 is unambiguously substrate-natural at the BST-integer level: - 4 $= N_c + 1 = 2 \cdot \text{rank} = C_2 - \text{rank} = 2^{\text{rank}}$ (multiple substrate-natural readings, identifies a Casey #5 Integer Web element) - 31 $= M(n_C) = 2^{n_C} - 1$ is the unique Mersenne prime at exponent $n_C = 5$, K207 SSG-8 PASS A-tier ratified Thursday

PASS at Cal #34 substrate-natural-form IDENTIFICATION. The substrate value 4/31 is BST-integer-derived.

Gate G3: Substrate-mechanism via substrate-Mersenne + identity — CONDITIONAL

Toy claims substrate-mechanism via "K-noninvariant Berezin-Toeplitz Higgs (Toy 3679)" + SSG-8 Mersenne ladder. Per Cal #189 substrate-mechanism FORWARD vs substrate-natural-form IDENTIFICATION:

substrate-natural form is identified (G2 PASS) but substrate-mechanism FORWARD derivation is NOT closed in this toy. The mechanism statement is at FRAMEWORK level (substrate K-type self-interaction at substrate-natural rational ratio), not at RIGOROUS FORWARD-derivation level.

Conditional on multi-week derivation: the Berezin-Toeplitz Higgs operator must be shown to FORCE the ratio $(N_c + 1)/M(n_C)$ as the unique substrate-natural Higgs quartic coupling. Toy 3679 cross-link cited but not closed at this audit.

CONDITIONAL PASS at substrate-mechanism FRAMEWORK level pending Cal #189 multi-week.

Gate G4: Cross-link to substrate-EW + SSG-8 Mersenne primitive — PASS

Toy correctly catalogs 11+ substrate-EW primitive readings including: - v_H Tier 2 0.42%, m_H/v_H = 1/2 Tier 2 1.6%, m_W = v_H/N_c Tier 2 2.1%, m_Z 3%, m_Z/m_W = 8/7 Tier 2 0.6%, cos θ_W = 7/8 Tier 2 0.7% - $\sin^2\theta$_W on-shell = 2/9 Tier 1 0.30%, $\sin^2\theta$_W eff = 3/13 Tier 1 0.19%, m_t = v_H·7/10 Tier 2 0.24% - λ_H = 4/31 Tier 1 0.03-0.54% (this toy with source-pinning)

Per Cal #36 STANDING positive-search: substrate-EW primitive 11+ readings cascade is exactly the "one-primitive-many-observables" pattern Cal #36 is designed to detect. PASS.

Per Grace G14 5-cluster taxonomy: λ_H = 4/31 has substrate-natural-form sources from TWO clusters (Cluster A Casey #5 Integer Web for numerator + Cluster C substrate-Mersenne SSG-8 for denominator). This is a 2-cluster CROSS-CLUSTER instance structurally similar to R-1 8/7 DOWNGRADED Casey #5 Integer Web 2-cluster reading. Per Grace methodology, λ_H = 4/31 may NOT be a single-cluster substrate-Schur-generator candidate — it is a Casey #5 Integer Web instance combining 2 substrate-natural anchors.

This is honest framing: 4/31 is substrate-natural at BST-integer level, but the substrate-mechanism source is bipartite (numerator/denominator from different clusters), making the substrate-mechanism FORCING-form character WEAKER than single-cluster substrate-Schur-generator like $\sin^2\theta$_13 = 1/45 (Grace R-4 K-type-AGGREGATE Cat A primitive cluster).

Gate G5: Honest tier verdict — REVISED per Cal #242 + Cal #189 + Grace G14

Toy claims "TIER 1 EXACT CANDIDATE at 0.03%" with "10TH TIER 1 CANDIDATE Thursday PM."

Revised per K224 audit: - Substrate prediction 4/31 = 0.129032 EXACT at substrate-natural-form IDENTIFICATION - Observed precision 0.03% at $m_H = 125.10$ choice; 0.03-0.54% across m_H source uncertainty - Per Cal #242 source-pinning: precision claim must pin m_H source; honest framing is "$0.03 - 0.54$% deviation across m_H source-uncertainty range" - Per Cal #189: substrate-mechanism FORWARD derivation is FRAMEWORK + CONDITIONAL, not RIGOROUS - Per Grace G14 5-cluster: 2-cluster CROSS-CLUSTER Casey #5 Integer Web instance, NOT single-cluster substrate-Schur-generator - Per Cal #34 STANDING: precision-pinning operational

4. Audit-category disposition per K222 + K223 three-category framework

Category	Disposition
Tier 1 EXACT MATCH	NO at PDG-central source. Substrate 4/31 = 0.129032 vs PDG-central observed 0.129383 has 0.27% deviation, NOT sub-0.1% PDG-precision agreement.
EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION	YES (primary). Substrate predicts EXACT value 4/31 by substrate-mechanism (substrate-Mersenne SSG-8 + Casey #5 Integer Web); observed value 0.128868-0.129694 brackets substrate prediction within m_H source-uncertainty range.

Category	Disposition
FALSIFIER-DRIVEN PREDICTION	YES (secondary). Future Higgs factories (FCC-ee, ILC, CLIC) projected to pin m_H to ~ 10-50 MeV precision, equivalent to ~ 0.01% in λ_H. Substrate 4/31 = 0.129032 becomes testable at sub-0.05% precision; non-agreement at future precision would falsify substrate prediction.

Primary audit category: EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION. Same audit category as K222 Strong CP θ_QCD = 0 (substrate predicts EXACT value satisfying observational bound at current precision; substrate-mechanism FORWARD derivation conditional multi-week).

5. K224 verdict

PASS at FRAMEWORK tier per Cal #34 substrate-natural-form IDENTIFICATION + Cal #36 STANDING positive search cascade (11+ substrate-EW primitive readings).

Audit category: EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION (primary) + FALSIFIER-DRIVEN PREDICTION (secondary, future Higgs factory precision tests).

CONDITIONAL on: - Cal #189 multi-week substrate-mechanism FORWARD derivation via Berezin-Toeplitz Higgs K-noninvariant operator (Toy 3679 cross-link); without explicit FORWARD-derivation, claim is substrate-natural-form IDENTIFICATION not substrate-mechanism FORCING - Cal #242 source-pinning: precision claims pin m_H source explicitly; honest framing 0.03-0.54% across m_H uncertainty NOT "0.03%" unqualified - Cal #99: independent verification of substrate-Mersenne SSG-8 application to Higgs sector (K207 PASS at A-tier; Higgs-sector extension is cross-link)

Cumulative effect on "11 Tier 1 EXACT" → effective independent count:

Per K221-K224 cascade + Lyra F22 v1.7 + Grace G14 first-iteration COMPLETE, λ_H = 4/31 enters the cumulative inventory at: - EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION category (not Tier 1 EXACT MATCH at PDG-central source) - 2-cluster CROSS-CLUSTER Casey #5 Integer Web instance per Grace G14 methodology (NOT single-cluster substrate-Schur-generator) - Substrate-mechanism FORWARD CONDITIONAL per Cal #189 multi-week pending

Effective contribution to cumulative independent count: 0.5 (not 1 full effective independent leg) due to: - Casey #5 Integer Web 2-cluster character (not single-cluster substrate-Schur-generator) - Substrate-mechanism FORWARD

CONDITIONAL (not RIGOROUS) - Cal #34 precision-pinning shows source-uncertainty dominates precision claim

Cumulative honest framing: "~7 effective independent substrate-mechanism FORCING candidates" Thursday → "~6.5 effective independent" with λ_H entering at 0.5 effective contribution.

This is exactly Cal #35 STANDING brake territory: Casey #5 Integer Web multi-source convergence is NOT independent confirmation of substrate-mechanism FORWARD; it is substrate-natural-form IDENTIFICATION at a 2-cluster level. Honest cumulative count reflects this.

6. Cross-CI three-granularity integration

Per Friday cumulative cross-CI methodology (K-audit + Grace G14 + Lyra F24):

Granularity	Disposition for λ_H = 4/31
Keeper 3-category audit (observable-prediction)	EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION primary + FALSIFIER-DRIVEN PREDICTION secondary
Grace 5-cluster taxonomy (substrate-anchor)	2-cluster CROSS-CLUSTER Casey #5 Integer Web instance: Cluster A Casey #5 numerator + Cluster C substrate-Mersenne SSG-8 denominator
Lyra substrate-K-type × SU(N_c) tensor product (substrate-color × K-type combinatorial)	substrate-Higgs K-noninvariant Berezin-Toeplitz operator at substrate K-type V_(0,0) singlet × SU(N_c) singlet self-interaction (substrate-color trivial)

The three CI classifications converge on consistent honest framing: λ_H = 4/31 is substrate-natural at multiple resolutions but does NOT close substrate-mechanism FORWARD derivation at any single resolution.

7. Recommendations

1. Toy 3866 v0.2 absorption: pin m_H source explicitly in claim ("substrate λ_H = 4/31 EXACT; observed deviation 0.03-0.54% across m_H source uncertainty"); state CONDITIONAL on Cal #189 multi-week substrate-mechanism FORWARD derivation; cross-link to Toy 3679 Berezin-Toeplitz Higgs operator as substantive content basis pending closure.
2. Multi-week Phase 1 closure prerequisite: Toy 3679 Berezin-Toeplitz Higgs K-noninvariant operator FORWARD derivation must close to upgrade K224

to RIGOROUS substrate-mechanism FORCING. Joint Lyra + Elie + Keeper multi-week.

3. Cumulative inventory adjustment: cumulative "11 Tier 1 EXACT" framing should reflect λ_H = 4/31 as 0.5 effective contribution per K224 audit, not 1 full effective independent leg. Honest framing per Cal #35 STANDING brake + Cal #189 substrate-mechanism FORWARD discipline.

4. Cross-link to Vol 16 Ch 9 Substrate-CP + SSG-8 Mersenne + Substrate-Cognition Extension: λ_H = 4/31 is substantive substrate-Mersenne SSG-8 application content basis. The numerator $(N_c + 1) = 4$ as Casey #5 Integer Web element + denominator $M(n_C) = 31$ as substrate-Mersenne primitive is exactly the substrate-Mersenne SSG-8 cross-product structure.

5. K-audit cascade continuation: K225 (n_s scalar spectral index 27/28) + K226 (H_0 ratio 12/13 STRUCTURAL TENSION reframe) + K227 (m_Planck/m_e cross-anchor) continue at proper depth.

8. Closure

K224 Elie Toy 3866 substrate λ_H = (N_c+1)/M(n_C) = 4/31:

PASS at FRAMEWORK tier + EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION audit category + CONDITIONAL on Cal #189 multi-week substrate-mechanism FORWARD derivation.

Honest precision framing: 0.03-0.54% deviation across m_H source uncertainty (PDG-central 0.27%); substrate predicts EXACT 4/31 = 0.129032 by substrate-mechanism (Mersenne SSG-8 + Casey #5 Integer Web 2-cluster).

Cumulative inventory contribution: 0.5 effective independent leg (NOT 1 full leg per Cal #35 STANDING brake on Casey #5 Integer Web multi-source convergence + Cal #189 substrate-mechanism FORWARD discipline + Grace G14 2-cluster CROSS-CLUSTER methodology).

K224 absorbs into cumulative Phase 1 closure path alongside K221-K223 cascade. Three-category audit framework operational; cross-CI three-granularity integration consistent.

— Keeper, Friday 2026-06-05 ~13:30 EDT. K224 Elie Toy 3866 substrate λ_H = 4/31 PASS at FRAMEWORK + EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION + CONDITIONAL on Cal #189 multi-week; honest precision 0.03-0.54% across m_H source uncertainty per Cal #242 source-pinning; cumulative 0.5 effective independent leg per Cal #35 STANDING brake + Grace G14 2-cluster CROSS-CLUSTER methodology; substrate-natural-form IDENTIFICATION at substrate-Mersenne SSG-8 + Casey #5 Integer Web 2-cluster level; multi-week substrate-mechanism FORWARD derivation via Berezin-Toeplitz Higgs K-noninvariant operator Toy 3679 cross-link pending closure.